

Dear Newcastle University CloseNIT team and Imperial Bioelectronics team,

I am writing to express my interest in the position of research assistant in closed-loop stimulation. My research journey has been shaped by a project that shares similarities with this topic, where I explored the application of auditory biofeedback with electrophysiology signals. The challenge of implementing this technology led me to delve deeper into neuroscience field, and throughout further specialised studies, I acquired advanced knowledge of fundamental neuroscience mechanisms and gained expertise in computational tools for analysing neural data and applying neural interfaces.

I believe my background aligns well with the requirements of this position, and I am eager to further develop my research skills with the aim of pursuing a PhD in the near future. I hold a BSc degree in Physics with Medical Physics from University College London and a MSc degree in Biomedical Engineering (Neurotechnology specialised) from Imperial College London. My academic journey has been multidisciplinary, across physics, biology, engineering and computing. This diverse educational background has provided me with a holistic understanding of implementing technology in real-world scenarios.

My previous experience with electrophysiology signal acquisition and auditory nervous system stimulation during my first individual project has equipped me with basic knowledge for this role. The process of designing signal filtering circuits, processing and analysing electromyography (EMG), and testing the prototype provided me with first-hand experience in developing a research idea into a tangible solution. Furthermore, through my Master's studies, I gained experience in decoding neural spike train, modelling neural network using programming languages. These experiences not only imparted transferable technical skills but also fostered critical thinking and problem-solving abilities.

Additionally, I have worked in an ultrasound lab as a research assistant, where I was entrusted with decision-making responsibilities and oversight of research progress. Regular discussions and reporting with both supervisors and research team members enhanced my communication and project management skills. Immersed in a pure research environment, I have practiced my ability to navigate complex research questions and foster productive collaboration.

I am excited about the possibility of contributing to your esteemed teams and drawing inspiration from cutting-edge research discoveries. Thank you for considering my application and I am looking forward to the opportunity to further discuss how my experiences align with this work position.

Sincerely,

Chuqi Wang